



Review paper

The future of pharmaceuticals: Artificial intelligence in drug discovery and development

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ABSTRACT

Artificial intelligence (AI) is revolutionizing traditional drug discovery and development models by seamlessly integrating data, computational power, and algorithms. This synergy enhances the efficiency, accuracy, and success rates of drug research, shortens development timelines, and reduces costs. Coupled with machine learning (ML) and deep learning (DL), AI has demonstrated significant advancements across various domains, including drug characterization, target discovery and validation, small molecule drug design, and the acceleration of clinical trials. Through molecular generation techniques, AI facilitates the creation of novel drug molecules, predicting their properties and activities, while virtual screening (VS) optimizes drug candidates. Additionally, AI enhances clinical trial efficiency by predicting outcomes, designing trials, and enabling drug repositioning. However, AI's application in drug development faces challenges, including the need for robust data-sharing mechanisms and the establishment of more comprehensive intellectual property protections for algorithms. AI-driven pharmaceutical companies must also integrate biological sciences and algorithms effectively, ensuring the successful fusion of wet and dry laboratory experiments. Despite these challenges, the potential of AI in drug development remains undeniable. As AI technology evolves and these barriers are addressed, AI-driven therapeutics are poised for a broader and more impactful future in the pharmaceutical industry.

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1. Introduction

Artificial intelligence (AI), defined as the intelligence demonstrated by human-made machines, emerged as a novel field of study dedicated to developing theories, methods, technologies, and applications aimed at simulating, extending, and enhancing human intelligence [1]. Over the past six decades, AI has evolved from a theoretical concept into a powerful industrial tool, revolutionizing industries such as manufacturing, agriculture, healthcare, and finance [2–5]. AI technologies have been successfully applied in areas including autonomous driving, voice recognition, web search, and medical diagnosis [6,7]. Its capabilities in specialized tasks like language translation and facial recognition now rival or surpass human performance, leading to the observation that “no field is immune to the charms and sweep of AI”. A watershed moment in

AI's history occurred in March 2016 when AlphaGo, an AI program, triumphed over renowned South Korean Go player Lee Sedol, sparking widespread societal debate [8]. The 2024 Nobel Prize in Physics went to two scientists, John J. Hopfield and Geoffrey E. Hinton, for foundational discoveries and inventions that enable machine learning with artificial neural networks. Furthermore, the Nobel Prize in Chemistry recognized using AI to design proteins. Proteins are the workhorse molecules of life, with millions existing in nature, but novel ones could transform medicine and technology. The new tools have already enabled researchers to churn out designer proteins for vaccines and cancer treatment, artificial pollution-eating enzymes, and molecular assemblies capable of seeding the growth of minerals [9,10].

2. The history of the development of AI drugs

AI has been applied in the pharmaceutical field for nearly three decades, with significant advancements since the late 1990s in its underlying algorithmic frameworks. Over this period, AI has undergone periods of rapid progress and setbacks [11], driven by the evolution from neural networks to deep neural networks (DNNs)

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and from ML to DL. Continuous optimization of algorithms, coupled with growing data accumulation and computational power, has been instrumental in advancing the AI field [12,13]. To evaluate the absorption, distribution, metabolism, excretion, and toxicity (ADMET) properties of new molecular entities (NMEs) as early as possible, various *in vitro* and *in vivo* methods including medium- and high-throughput screening have been developed, which also facilitate the rapid accumulation of experimental data. However, as the number of NMEs continues to increase, these experimental approaches have shown several inherent shortcomings: time-consuming, costly, and animal welfare issues involved, which have greatly limited their application and spurred the emergence of *in silico* methods for predicting ADMET properties. In recent decades, with the rapid development of computer science and the accumulation of ADMET experimental data, *in silico* predictive models and derived web tools aimed at facilitating the efficient evaluation of ADMET properties have been greatly developed [14].

Since 2018, AI in pharmaceuticals has advanced from a conceptual phase ("0") to practical application ("1") (Fig. 1). Although no AI-enabled drugs have been approved by the U.S. Food and Drug Administration (U.S. FDA) for marketing yet, several AI-driven pharmaceutical companies have successfully accelerated phases I and II clinical candidates [15]. In 2024, recent research highlighted a breakthrough in drug design using DL to reverse-engineer synthetic routes, an achievement compared to AlphaGo's impact on chemistry [16,17]. This marked the beginning of significant breakthroughs in AI-driven pharmaceuticals. For instance, in 2021, Healex utilized AI to identify new uses for the drug HLX-0201 in treating fragile X syndrome, advancing the project to phase II clinical trials within 18 months [18]. In 2019, Deep Genomics applied its AI platform to identify novel targets and screen oligonucleotide candidates for Wilson's disease, completing the process in just 18 months [19]. Insilico Intelligence (Insilico Medicine) utilized GENTRL, a generative adversarial network (GAN)-based approach, to complete an AI drug discovery challenge within 21 days. This process involved data collection, model development, and the design of novel molecules, ultimately generating a highly active discoidin domain receptor 1 (DDR1) kinase inhibitor [20]. Although the identified compounds demonstrated satisfactory microsomal stability and pharmacokinetic properties, further optimization is required to improve selectivity, specificity, and other critical

medicinal chemistry parameters. Additionally, DeepMind's AlphaFold 3 achieved a significant breakthrough in addressing a 50-year-old biological challenge by accurately predicting the three-dimensional (3D) structure of proteins [21]. In March 2024, InSys Intelligence's fully AI-generated drug for idiopathic pulmonary fibrosis (IPF) entered phase IIa trials. This drug, with a novel backbone compound developed by Chemistry42 using AI software Pandaomics, showcases AI's potential in innovative drug development [22]. However, it is crucial to recognize the limitations of AI in drug discovery. Analyses derived from multiple AI methods may be misleading due to issues like overlap between testing and training datasets, biases in the data, or a lack of chemical insight into the results. These biases can produce high apparent accuracy, but often with poor generalizability and limited applicability in prospective research.

Nonetheless, AI's role in predicting and screening new therapeutic targets and drugs remains a highly promising area of research [23]. Over the years, AI has been consistently heralded as a transformative tool in accelerating drug discovery, development, and testing, significantly reducing research timelines. Presently, numerous AI-enabled drug development pipelines are entering clinical phases globally (Table 1) [24–64].

3. AI pharmaceutical elements

3.1. The elements of AI

The three core components of AI, i.e., data, computation, and algorithms, serve as the foundation of AI-driven pharmaceutical research. Data sources in the pharmaceutical sector include public and commercial datasets, research and development (R&D) data obtained through collaborations with pharmaceutical companies, in-house research datasets, and those generated through data mining and manual cleaning and validation. Advancements in computational power, particularly through graphics processing unit (GPU) cloud computing resources, have provided critical support for AI pharmaceutical companies [65]. Different algorithmic models are tailored to specific application scenarios, and when combined with unique data sources, they give rise to the distinctive profiles of AI companies [66].

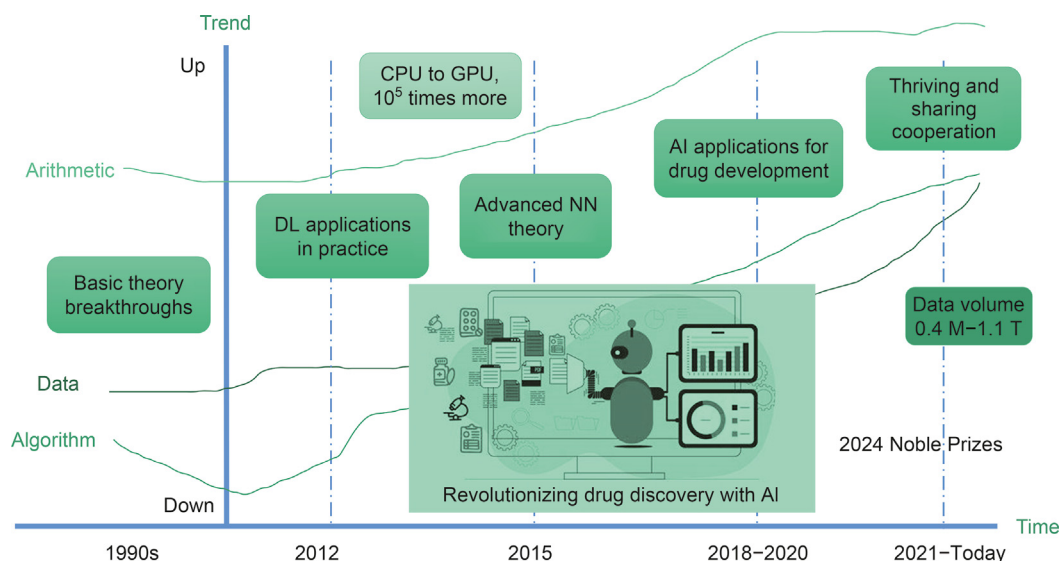


Fig. 1. Brief overview of artificial intelligence (AI) pharmaceutical development. DL: deep learning; CPU: central processing unit; GPU: graphics processing unit; NN: neural network.

Table 1
Artificial Intelligence (AI)-enabled drugs entering the clinical phase.

No.	Company	Pipeline	Indications	Clinical phase	Clinical trials No.	Update year	Refs.
1	Recursion	REC4881	Familial adenomatous polyposis	Phase I	NCT05552755	2025	[24]
2		REC2282	Neurofibromatosis type 2	Phase II/III	NCT05130866	2024	[25]
3		REC994	Cerebral cavernous malformation	Phase II	NCT05085561	2024	[26]
4	Lantern	LP100	mCRPC	Phase II	NCT03643107	2025	[27]
5		LP300	Lung adenocarcinoma	Phase II	NCT05456256	2025	[28]
6		LP184	Solid tumors	Phase I	NCT05933265	2025	[29]
7	Relay	RLY1971	SHP2	Phase I	NCT04252339	2023	[30]
8		RLY4008	FGFR2	Phase I	NCT04526106	2025	[31]
9		RLY2608	PI3K α	Phase I	NCT05216432	2025	[32]
10	Accutar Biotechnology	AC682	Breast cancer	Terminated	NCT05489679	2024	[33]
11		AC176	mCRPC	Phase I	NCT05241613	2025	[34]
12	Berg Health	BPM31510	Glioblastoma	Phase II	NCT04752813	2025	[35]
13		BPM31543	Alopecia	Phase I	NCT01588522	2017	[36]
14	AI Therapeutics	LAM-001	BOS and pulmonary sarcoidosis	Phase II	NCT05798923	2024	[37]
15		LAM-002A	Amyotrophic lateral sclerosis	Phase II	NCT05163886	2024	[38]
16	Benevolent AI	BEN-2293	Atopic dermatitis	Phase II	NCT04737304	2023	[39]
17	BioXcel Therapeutics	BXCL501	Acute agitation	Phase II	NCT05276830	2023	[40]
18		BXCL701	Metastatic castration-resistant prostate cancer	Phase II	NCT03910660	2023	[41]
19	Exscientia	EXS21546	Oncology	Phase I	NCT04727138	2022	[42]
20	Evaxion Biotech	EXV-01	Metastatic melanoma	Phase II	NCT05309421	2023	[43]
21		EXV-02	Adjuvant melanoma	Phase I/II	NCT04455503	2024	[44]
22	Pharos Ibio	PHI-101-001	Acute myelogenous leukemia	Phase I	NCT04842370	2021	[45]
23		PHI-101-002	Platinum-resistant ovarian cancer	Phase I	NCT04678102	2023	[46]
24	SOM Biotech	SOM0226	Familial amyloid polyneuropathy	Phase II	NCT02191826	2016	[47]
25		SOM3355	Huntington chorea	Phase II	NCT05475483	2024	[48]
27		SOM0061	COVID-19	Phase II	—	2024	[49]
28	Neumora	BTRX-335140	Major depressive disorder and anhedonia	Phase II	NCT04221230	2023	[50]
30	Xbiome	XBI-302	Acute-graft-versus-host disease	Phase I	NCT05352269	2022	[51]
31	Erasca	ERAS-007	Metastatic colorectal cancer	Phase I/II	NCT05039177	2024	[52]
32		ERAS-601	Advanced or metastatic solid tumors	Phase II	NCT04866134	2024	[53]
33	Nimbus Therapeutics	NDI-034858	Psoriatic arthritis	Phase II	NCT05153148	2024	[54]
34		NDI 1150-101	Solid tumor	Phase I/II	NCT05128487	2024	[55]
35	Landos Biopharma	BT-11	Crohn's disease	Phase II	NCT03870334	2023	[56]
36		NX-13	Ulcerative colitis	Phase I	NCT04862741	2023	[57]
37	C4X discovery	INDV-2000	Orexin-1	Phase I	NCT05694533	2024	[58]
38	Oncocross	OC514	Sarcopenia	Phase I	NCT05264038	2023	[59]
39	Pharnext	PXT3003	CMT1A	Phase II	NCT05092841	2025	[60]
40	Healx	HLX-0201	Fragile X syndrome	Phase II	NCT04823052	2022	[61]
41	AbCellera	LY-CoV555	COVID-19	Completed	NCT05780268	2023	[62]
42	Insilico Medicine	INS018-055	IPF	Phase I	NCT05154240	2023	[63]
43	Schrodinger	SGR-1505	IPF	Phase I	NCT05544019	2024	[64]

mCRPC: metastatic castration-resistant prostate cancer; SHP2: Src homology 2-containing protein tyrosine phosphatase 2; FGFR2: fibroblast growth factor receptor 2; PI3K α : phosphatidylinositol 3-kinase α ; BOS: bronchiolitis obliterans syndrome; COVID-19: coronavirus disease; CMT1A: charcot-marie-tooth disease, type 1A; IPF: idiopathic pulmonary fibrosis.

3.2. AI, ML, and DL

In the 2020 CASP14 competition, DeepMind's AlphaFold 2 achieved a groundbreaking advance in protein structure prediction, outpacing the second-place competitor by a substantial margin [67]. Media outlets widely referred to this achievement using terms such as AI, ML, and DL [68]. AI encompasses machine/computer vision and NLP agents capable of perceiving their environment and reacting to it to achieve specific objectives (Fig. 2). The fundamental approach involves “training” machines with algorithms and data to enable them to perform tasks and make predictions or inferences about future outcomes. The industry often classifies ML algorithms in two ways: based on learning scenarios or by their form and function [69]. DL, a more advanced form of ML, utilizes combinatorial non-linear models that automatically learn effective features at multiple levels from high-dimensional, complex data [70]. The “depth” in DL refers to the number of layers in the network; more layers indicate a deeper network. Each module within the network transforms its input into higher-level, more abstract representations [71]. DL models are often considered a form of end-to-end learning, where the learning process is integrated without separating it into discrete modules or stages. Instead, “input-output” pairs are provided as training data, which the system optimizes

throughout the process to achieve the task without propagating errors common in traditional ML [72]. Currently, DL is predominantly based on neural network models, which feature a multi-layered structure designed to mimic the human brain and learn from large data volumes [73]. These systems have demonstrated superior accuracy and sophistication compared to other ML methods, proving highly successful in tasks such as image and sound recognition [74]. DL has catalyzed AI's rapid growth over the past decade, with applications already deeply embedded in computer vision and NLP. Furthermore, DL is emerging as a powerful tool in complex fields like high-energy physics, computational chemistry, computational biology, and medical diagnostics, where expert knowledge is essential [75].

Previous studies have demonstrated that DL technology offers significant advantages in optimizing chemical synthesis routes, predicting drug pharmacokinetics, identifying drug target sites, and generating new molecular structures [76]. DL models learn the intrinsic relationships between compounds and target proteins by training on extensive datasets of known compound-target protein interactions. This training enables DL models to automatically extract relevant features from both compounds and target proteins, as well as discern interaction patterns. For example, Guttman and Kerem [77] developed a cytochrome P-450 3A4 (CYP3A4) inhibitor

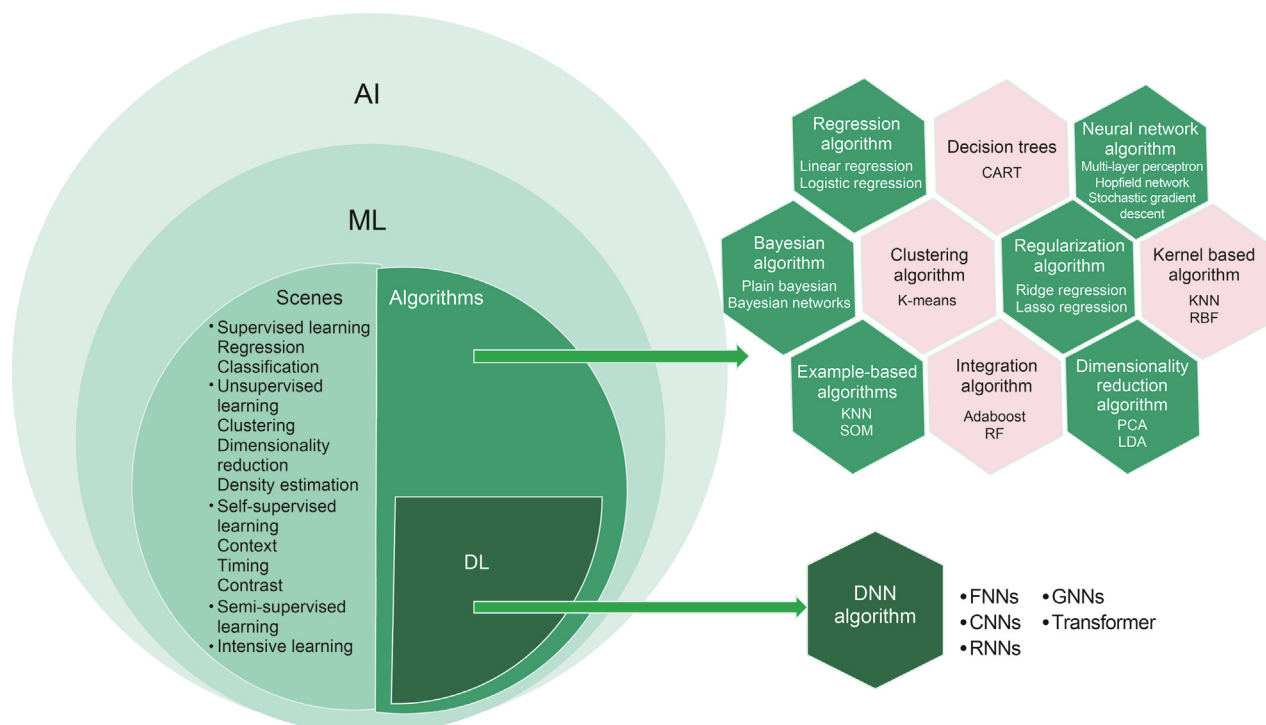


Fig. 2. Overview of artificial intelligence (AI), machine learning (ML), and deep learning (DL). CART: classification and regression tree; KNN: K-nearest neighbor; RBF: radial basis function; SOM: self-organizing maps; RF: random forest; PCA: principal component analysis; LDA: linear discriminant analysis; DNN: deep neural network; FNNs: feedforward neural networks; GNNs: graph neural networks; CNNs: convolutional neural networks; RNNs: recurrent neural networks.

prediction model using the DeepChem framework. Since the introduction of the Lipinski rule of five for drug design, predicting the early ADMET properties of lead compounds has gained increasing importance. Several studies have shown that training DL models on large datasets of known compounds' ADMET properties allows for the automatic identification of relationships between compound characteristics and their properties. Well-trained DL models can then predict the properties of novel compounds, thereby accelerating drug discovery and development. In recent years, AI has begun to revolutionize chemical synthesis. However, the lack of suitable chemical reaction characterizations and the scarcity of reaction data have limited the widespread application of AI in reaction prediction. DL can address these challenges by automatically identifying and extracting patterns from chemical synthesis routes, predicting the efficiency and selectivity of new synthetic routes, and accelerating the development of new drugs by analyzing large datasets of chemical synthesis reactions.

The following outlines several applications of DL algorithms in virtual screening (VS): convolutional neural networks (CNNs) are particularly effective for processing image data such as molecular structure diagrams. By identifying and extracting features within molecules, such as atom types, positions, and chemical bonds, CNNs can predict the properties and activities of molecules. For VS tasks involving sequence data (e.g., chemical molecular sequences), recurrent neural networks (RNNs) are well-suited. RNNs excel at capturing long-term dependencies within molecular sequences, improving the accuracy of property predictions. GANs are instrumental in generating novel molecular structures, a key advantage in VS. By training GANs, it is possible to generate molecules with desired properties, significantly reducing the need for experimental validation. Graph neural networks (GNNs) are ideal for processing graph-structured data, such as molecular graphs [78]. GNNs can model the relationships between atoms and chemical bonds, facilitating more accurate predictions of molecular properties. For

tasks involving long sequence data, such as multi-step chemical reaction prediction, Transformer models are particularly effective. Transformers can capture long-term dependencies within sequences, providing enhanced accuracy in predicting molecular properties [79].

4. AI in the pharmaceutical

Drug discovery has historically relied heavily on serendipity, with many significant breakthroughs occurring through chance observations or unintended findings [80]. However, AI offers the potential to remove much of the uncertainty in this process, dramatically improving the chances of identifying commercially viable drug candidates while reducing both costs and time. A predictive study in 2022 concluded that by heavily investing in AI, the pharmaceutical industry could see a return on investment increase of more than 45% [81]. The drug development process, which aims to identify biologically active compounds for disease treatment, typically begins with the identification of molecular targets, followed by the discovery of active drug candidates, and progresses to the optimization of lead compounds for preclinical and clinical trials, ultimately leading to regulatory approval. This process is not only time-consuming but also high-risk and expensive, with the average cost of developing a new drug ranging between \$100 million and \$2 billion, and the timeline stretching from 10 to 17 years. Even if a drug candidate successfully passes phase I clinical trials, it has only a 5% chance of reaching the market [82].

Before 1980, drug discovery primarily relied on random screening and empirical observations of natural product effects on known diseases [83]. Although inefficient, this method led to the discovery of several groundbreaking drugs, such as penicillin in the 1940s, which revolutionized the treatment of previously incurable diseases like tuberculosis and bacterial infections. Other notable discoveries include antihypertensive drugs like prilosec, lipid-

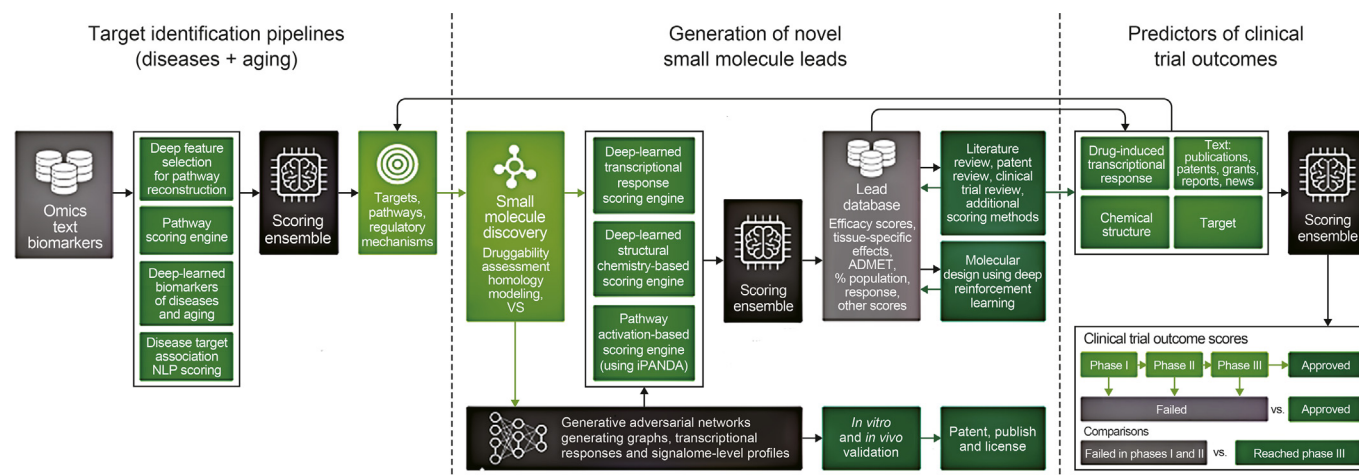


Fig. 3. Artificial intelligence (AI) for the pharmaceutical industry. VS: virtual screening; ADMET: absorption, distribution, metabolism, excretion, and toxicity; PK: pharmacokinetics.

lowering statins, and anticoagulants like clopidogrel, which played a pivotal role in managing cardiovascular diseases. However, after the 1980s, the drug discovery process improved significantly with the advent of high-throughput screening (HTS), which automates the testing of thousands of compounds against molecular targets or cellular assays. A notable milestone in HTS was the discovery of the immunosuppressant cyclosporine A in 1988 [84]. In parallel, researchers continued to invest in novel methods to enhance the drug discovery process. Computer-aided drug design (CADD) emerged as an effective approach to streamline the development of new drugs [85]. CADD utilizes molecular modeling techniques to analyze the structure and interactions of numerous molecules, both solid and non-solid, against pharmacological targets, assessing their activity, toxicity, and bioavailability. This facilitates better planning and guidance throughout the drug discovery process [86] (Fig. 3). Several significant drugs, including the anti-hypertensive drug captopril, the anti-human immunodeficiency virus (HIV) drugs saquinavir, zidovudine, and zalcitabine, and the protease inhibitor boceprevir for hepatitis C, have been discovered using VS techniques. In the past decade, AI has gained increasing prominence in CADD, yielding more accurate predictive models. The emerging field of AI drug design (AIDD) is now widely recognized within the pharmaceutical industry. It is anticipated that the integration of AI in CADD will continue to revolutionize drug R&D, making future drug discovery efforts faster, more cost-effective, and more successful [87,88].

4.1. Drug characterisation

The encoding of molecules as fixed-length strings or vectors is a prerequisite for AI-driven drug molecule research [89]. Given the vast chemical space of drug molecules, selecting appropriate molecular features to accomplish specific tasks is essential. Molecular characterization, also referred to as molecular descriptors, plays a pivotal role in accurately modeling and predicting the properties and biological activity of small molecules. Such characterizations are essential for applications in virtual drug screening, compound search, ADME/T prediction, inverse synthetic route planning, and other drug discovery processes [90,91].

The article discusses various types of molecular fingerprints (MFPs), such as simplified molecular input line entry system (SMILES), substructure-based, hash-based, and pharmacophore-based fingerprints. A comparative analysis of these methods'

performance metrics (e.g., accuracy, speed, and memory usage) on benchmark datasets would provide valuable guidance for selecting the most suitable approach for specific research needs.

Regarding GNNs for drug property prediction, the article briefly highlights their potential. Evaluating the performance of different GNN architectures (e.g., graph convolutional network (GCN), graph attention network (GAT), and Graph Sample and aggregate (GraphSAGE)) on benchmark datasets, alongside an assessment of their interpretability and computational efficiency, would offer critical insights into the advantages and limitations of each model.

4.1.1. SMILES

SMILES is widely employed for drug characterization, encoding molecular structures and geometric properties. Its linear molecular representation allows SMILES strings to be processed directly as text, making it especially useful in DL models for various drug design tasks, such as inverse synthesis prediction via sequence-to-sequence (seq2seq) methods. Additionally, SMILES' ability to generate multiple representations of the same molecule by altering atomic order provides an advantage for data augmentation [92].

4.1.2. Molecular fingerprinting

A MFP is a bit string that encodes the structural or pharmacological properties of a molecule [93]. MFPs are widely utilized in ligand-based similarity searches and quantitative structure-activity relationship (QSAR) analysis, especially in VS for drug discovery. Additionally, DL-based drug-target interaction (DTI) prediction models frequently use MFPs as input features [94]. DTI prediction assists researchers in assessing the efficacy and safety of drug candidates early in the process, narrowing the search for therapeutic targets, and expediting the discovery and development of new drugs [95].

Drugs and targets are typically represented as 1D sequences, with DL models (such as CNNs, RNNs, and Transformers) employed to extract features and make predictions. These models are designed to effectively represent the complex structures of drugs and targets, capturing their interactions. They also aim to enhance the interpretability of predictions, provide insights into the model's internal workings, and ensure the model generalizes well to unseen data. To address these challenges, new DL models such as mutual Transformer-drug target affinity (MT-DTA), multi-scale diffusion and interactive learning-drug target affinity (MDCT-DTA), and Transformer-graph drug-target affinity prediction (TGraphDTA)

have been proposed [96,97]. These models combine techniques like diffusion models, graph optimization, and interaction learning to improve feature characterization and prediction accuracy. Furthermore, the interpretability of these models has been enhanced by visualizing key molecular structures upon which the models focus. Efforts to improve feature characterization include combining the 3D structures of proteins with drug complex structures, as well as developing more interpretable models using explainable AI technologies. To improve model generalization, larger-scale biochemical datasets are being incorporated [98]. Common molecular fingerprinting methods include substructure-based, hash-based, and pharmacophore-based fingerprints. Notable substructure-based MFPs include the molecular access system (MAS) and PubChem fingerprints, which are used for neighborhood and similarity searches. The PubChemFP encodes 881 structural key types corresponding to the substructures of all compound fragments in the PubChem database. Hash-based fingerprints, such as Daylight FP, Morgan FP, and extended connectivity fingerprints (ECFPs), are widely used for compound similarity analysis [99]. Unlike substructure-based methods, hash fingerprints convert all possible fragments into values using a hash function. Among these, ECFP, a recurrent fingerprint based on Morgan's algorithm, is commonly used as input for DNNs in bioactivity prediction and has shown strong stability [100]. Pharmacophore fingerprints assign pharmacophore types to atoms in the chemical structure, generate multiple conformations, and construct binary fingerprints based on these pharmacophores. These fingerprints are used as descriptors in partial least squares QSAR models. By capturing molecular features such as aromaticity, hydrophobicity, charge, and hydrogen bond donor/acceptor properties, pharmacophore fingerprints enable the assessment of similarity between target binding sites, considering energy-minimized conformations of molecules to extract key pharmacophore features [101].

4.1.3. Molecular characterisation learning

The two methods of molecular characterization described above yield numerous molecular descriptors; however, the results are often constrained by the domain-specific expertise of the computational chemist and the choice of algorithm employed [102]. Determining which molecular structures and properties to characterize for optimal downstream processing remains a challenge [103]. GNNs offer a comprehensive and generalized approach to molecular characterization. By representing atoms as graph nodes and chemical bonds as graph edges, molecular graphs are transformed from abstract mathematical concepts into concrete representations that can be processed by computers. These graphs are mapped onto linear data structures, such as matrices or arrays, facilitating computational handling [104]. In a GNN-based molecular graph, each atom and bond is associated with an initial feature vector within a feature matrix. The atom's feature vector typically includes information about its local chemical environment, such as atomic type, formal charge, and the number of attached hydrogens. Bond features may include the adjacency matrix, bond type, shortest path length, and the presence or absence of specific rings [105]. GNNs can automatically learn task-specific molecular representations through graph convolution, eliminating the need for traditional manual descriptors or fingerprints, and they demonstrate high accuracy in predicting compound properties [106]. Predicting a molecule's chemical properties or biological activity directly from its structure has long been a focus of interest within the chemical community [107]. The GNN-based fingerprinting method, Neural FP, utilizes graph CNNs to learn molecular representations directly from molecular graphs. The Weave model considers both atoms and chemical bonds within the molecular graph,

optimizing atomic and atomic pair features, and has proven effective in predicting water solubility, biological activity, and toxicity. Similarly, Attentive FP introduces a graph attention mechanism to model node information, capturing local and non-local features of chemical structures, such as intramolecular hydrogen bonds and aromatic systems. This enables Attentive FP to excel at learning molecular representations for a wide range of properties [108]. Beyond drug property prediction and characterization, GNNs are also applicable in areas such as *ab initio* drug design, interaction prediction, and inverse drug discovery [109].

4.2. Target discovery and validation

Targeted drug discovery remains a cornerstone of pharmaceutical development. When a drug's target is known, designing drug screening experiments to identify therapeutics acting on that protein target becomes more straightforward [110]. However, failing to identify a target accurately can result in significant R&D investment losses. For instance, clinical trials by Pfizer, Roche, and Merck Sharp & Dohme for cholesteryl ester transfer protein (CETP) inhibitors, a lipid-lowering target, ended in failure [111]. Conversely, the discovery of programmed cell death 1 (PD-1) has revolutionized biomolecule and tumor immunotherapy, and it is projected that more than 20 PD-1 products will reach the market globally within the next 2–3 years. On the other hand, even when a novel druggable protein target is discovered, the path to bringing a new chemical entity to market is fraught with significant challenges, especially concerning development time and cost. Identifying new targets or indications for an existing drug, however, can substantially reduce development expenses [112]. One of the most notable examples of drug repurposing is sildenafil, and AI technology has the potential to transform such serendipitous discoveries into more systematic successes [113]. By combining systems biology with AI algorithms, correlations between multi-omics data and patient clinical health information can be mined. Additionally, using NLP to retrieve and analyze unstructured data from literature, patents, and clinical reports can help uncover potential disease-relevant pathways, proteins, and mechanisms. This approach aids in the identification of new targets for drug development, whether for novel chemical entities or repurposed drugs [114,115].

i) Comparison of reverse docking software: the article mentions reverse docking as a method for target discovery. A performance comparison of various reverse docking software (e.g., AutoDock, Glide, and Rosetta) on a benchmark dataset, considering factors like ease of use and scalability, could help researchers identify the most suitable tool for targeting specific proteins.

ii) Comparison of protein structure prediction methods: the article discusses AlphaFold as a protein structure prediction method. Comparing its performance (e.g., Global Distance Test (GDT) score) with other methods such as I-TASSER or Rosetta on a benchmark dataset could provide valuable insights into the strengths and limitations of different protein structure prediction approaches.

4.2.1. Systems biology approach

By examining the interrelationships and interactions of various components within biological systems at the molecular level, such as gene and protein networks related to cell signaling, metabolic pathways, organelles, cells, physiological systems, and organisms, systems biology aims to create comprehensive models and complete organism maps [116]. Network-based approaches infer new protein phenotypes or associations by linking proteins/genes to different network pathways [117]. However, the intricate complexity of biological network interactions presents challenges in constructing network-based models for disease classification,

personalized medicine, and prognosis. These models often fail to provide stable pathway signatures for specific phenotypes or reliable biomarkers of disease, hindering the creation of unbiased, data-driven networks for identifying biomarkers, targets, and diseases [118].

Utilizing Bayesian AI analysis to integrate molecular profiles from multi-omics data (such as genomics, proteomics, lipidomics, and metabolomics) with clinical health information enables the construction of causal inference networks. By comparing the differences between "health" and "disease" network graphs, disease drivers (targets and biomarkers) can be identified. This approach led to the discovery of the novel tumor target BPM42522, its lead molecule, and its anticancer mechanism of action [119,120].

The integration of knowledge mapping techniques with systems biology to build biomedical knowledge graphs has increasingly become pivotal in medical practice and research. These graphs help to simplify complex biological systems and pathological processes, offering a clearer understanding of underlying principles. When combined with disease-specific contexts, biomedical knowledge graphs facilitate drug repurposing and mechanistic analysis of emerging human diseases, such as coronavirus disease 2019 (COVID-19) [121]. BenevolentAI has introduced a judgment-enhanced cognitive system (JECs) that uses AI tools and biomedical knowledge graphs to identify potential drug candidates. By discovering new connections between vast amounts of unstructured data, such as disease, drug, and clinical trial information, JECs enables drug redirection and assists scientists in identifying new indications for existing drugs [122]. Similarly, MindRankAI has developed PharmKG39, a multi-relational biomedical knowledge graph of drug-disease associations, which integrates over 500,000 relationships between genes, drugs, and diseases. Using a heterogeneous graph attention neural network, PharmKG39 incorporates 29 relationship categories and over 8000 ambiguous entities, each enriched with domain-specific information from datasets on gene expression, chemical structure, and disease word embedding, preserving both semantic and biomedical features [123].

The development of the *in silico* pathway activation network decomposition analysis (iPANDA) method by scientists at Insilico Medicine represents a significant advancement in pathway activation analysis [124]. iPANDA is designed to extract biologically relevant features from large-scale transcriptomic and proteomic data, offering a powerful approach for biomarker identification. The method uses gene expression data to assess fold changes between tumor samples and normal group averages, incorporating gene importance factors to quantify the influence of genes on specific pathways. However, the measure of gene centrality varies across algorithms, and this variation can lead to highly inconsistent results [125]. In this particular study, iPANDA integrates the degree of differential gene expression with pathway topology decomposition into a unified network model. By utilizing statistical and topological weights, gene importance is estimated. Additionally, gene co-expression modules are introduced, and the topological coefficients of these modules are computed to obtain co-expression data. This data is then combined with the gene importance factors to calculate pathway activation scores. Leveraging iPANDA, Insilico Medicine has developed a new target discovery platform called PandaOmics, which has been instrumental in identifying and prioritizing over 20 novel targets for IPF. The platform compares histology data from patients with fibrosis to healthy individuals, identifying significant differences and utilizing iPANDA technology to pinpoint pathways that may affect these differences. Following this, target safety and future value were assessed through target knockout data, leading to the identification of promising targets for IPF treatment. The project has progressed to clinical stages,

marking a significant milestone in the development of therapies for IPF [126].

4.2.2. Target structure-based approaches

Target confirmation (i.e., target selection or prioritization) in drug development remains an uncertain process, necessitating precise mapping of interactions between approved drugs and their efficacy targets, i.e., the specific proteins or molecules on which the drugs exert therapeutic effects [127]. Structure-based computational methods for target discovery serve as valuable complements to experimental strategies, such as reverse docking, pharmacophore modeling, binding site similarity, and fingerprint-based interactions [128]. Among these, reverse docking has proven particularly effective, not only for target validation but also for predicting toxicity and side effects, as well as for uncovering novel, previously unidentified targets for drugs or natural compounds [129]. The Potential Drug Target Database (PDTD), a comprehensive database for reverse docking, has been used to identify potential targets for compounds such as tea polyphenols and ginsenosides. However, the reverse docking method is constrained by the available target structure dataset. The PDTD, released in 2008, includes approximately 1100 protein entries with 3D structures, sourced from literature and various online repositories (e.g., Therapeutic Target Database (TTD), DrugBank, and Thomson Pharma), covering 830 known or potential drug targets [130]. Notably, only approximately 11% of the human proteome has been annotated with small molecule probes, leaving a significant portion of proteins, approximately one-third of the human proteome, still uncharacterized in terms of their biological functions and roles in disease.

Three methods, nuclear magnetic resonance (NMR), X-ray crystallography, and cryo-electron microscopy, are widely employed for protein structural resolution and have yielded significant insights into protein structures and drug receptors. Many drugs, such as angiotensin-converting enzyme (ACE) inhibitors, have entered clinical practice based on structural data derived from these techniques [131–133]. However, these methods are not without their challenges. Cryo-electron microscopy equipment is priced between \$20 million and \$60 million, while the synchrotron light sources required for X-ray crystallography can cost several hundred million dollars to construct [134]. Additionally, the time required to resolve a protein structure can range from weeks or months to several years, influenced by factors such as sample availability and protein complexity [135]. In light of these constraints, AI-based protein structure prediction algorithms present a promising complement to traditional protein information-driven target validation approaches. AlphaFold 2, for instance, achieved a median GDT score of 92.4 across all targets, closely approximating the quality of results from gold-standard experimental methods like X-ray crystallography [136]. AlphaFold 3, developed by DeepMind, employs a DNN architecture trained on 170,000 protein structures from the Protein Data Bank (PDB) to predict inter-amino acid distances and torsion angles between bonds in protein structures [137]. The methods and architecture behind AlphaFold 3 were recently published, and in collaboration with European Molecular Biology Laboratory-European Bioinformatics Institute (EMBL-EBI), AlphaFold 3's predictions, covering 98.5% of the human proteome, have been made publicly available for the scientific community [138]. Although AlphaFold 3 still struggles with accurately modeling side-chain structures and dynamics and faces challenges in predicting the structures of multi-domain proteins, protein complexes, and membrane proteins, its extensive protein structure library provides a nearly comprehensive reverse docking dataset. This resource enables the identification of potential target proteins, thus helping to mitigate the limitations of existing target structure datasets in reverse docking. As a result, reverse docking,

bolstered by AlphaFold 3's insights, has the potential to become a truly invaluable tool in drug discovery, advancing the field significantly [139–141].

4.3. Small molecule drug discovery

In recent years, DL has significantly impacted fields such as image analysis and NLP. Motivated by these advancements, computational chemists are increasingly employing generative models to design new molecules and predict their properties [142]. The vast chemical space encompasses approximately 10^{60} to 10^{100} possible small molecules, making drug discovery akin to finding a needle in a haystack, as researchers must identify compounds that satisfy multiple criteria, such as biological activity, metabolic stability, and potency. Consequently, only a tiny fraction of this theoretical chemical space can be explored through traditional experimental approaches. Computer modeling techniques, however, are proving invaluable in enhancing the biological screening of large compound libraries and optimizing synthetic routes to complementary compounds, thus playing a pivotal role in early drug discovery [143].

i) Comparison of molecular generator techniques: the article discusses the use of GANs for molecule generation. Comparing the performance of various GAN architectures, such as conditional generative adversarial network (CGAN) and style-based generator architecture for generative adversarial networks (StyleGAN), on benchmark datasets, and evaluating their ability to generate novel and biologically active molecules, could assist researchers in selecting the most effective technique for drug discovery purposes.

ii) Comparison of synthetic route planning methods: the article also addresses template-based and template-free approaches for synthetic route planning. A comparison of their performance, considering factors such as efficiency, selectivity, and by-product formation, on benchmark datasets, could guide researchers in choosing the most suitable strategy for synthesizing their target molecules.

4.3.1. Molecular generator techniques

A key aspect of compound design and predictive modeling is the selection of appropriate molecular representations. Text or string encoding of molecules is computationally inexpensive and commonly used in molecular generators. In generative modeling, SMILES-based string encoding typically generates a token for each atom, which is then converted into a "one-hot" string representation [144]. A generative model using one-hot encoding produces a distribution for each token, which is sampled to generate a new structure in SMILES format. Alternatively, graph-based generative modeling, using techniques like GCNs or DL to generate molecular structures, is an emerging area of research. Rule-based graph generative models often yield structurally correct molecules but are computationally intensive. The integration of flexible neural network architectures and diverse molecular representations has led to the development of various innovative approaches for molecular generation [145].

4.3.2. Synthetic route planning

Computer-aided synthetic planning (CASP) has its roots in the pioneering work of E.J. Corey, who formalized the concept of "inverse synthetic analysis" in the late 1960s [146]. CASP incorporates the principles of inverse synthetic analysis to help synthetic organic chemists identify the most efficient and cost-effective synthetic routes, predict selectivity and by-products, and suggest optimal reaction conditions. Over the decades, computational methods have evolved from expert systems based on manually coded reaction rules and templates to data-driven, AI-assisted synthesis

planning [147]. Modern AI algorithms are now available to recommend feasible synthetic routes for a wide range of reactions, with or without reaction templates, operating at either the mechanistic or global reaction level. These methods utilize molecular representations such as fingerprints, graphs, or even SMILES strings. CASP is instrumental in enabling chemists to make better decisions, thereby increasing efficiency and productivity, reducing synthetic failures, and accelerating the design-make-test-assess (DMTA) cycle in drug discovery [148].

Rule-based approaches rely on expertly coded rules and heuristics extracted from reaction databases and literature to suggest synthetic routes, often referred to as "template methods". In such approaches, reaction rules are manually curated, which is limited by the inability to scale with the exponential growth of chemical literature and by the finite knowledge base that cannot be fully comprehensive. Synthia (Chematica) is an inverse synthesis software that leverages a library of expertly coded rules for chemical synthesis planning [149]. To address the limitations of the rule-based system, Synthia incorporates computational methods to automate the extraction of reaction rules from reaction datasets. Its template extraction algorithm, based on Ambit-SMIRKS, is specifically designed to describe chemical reactions and has accumulated an expert-coded rule base of approximately 50,000 rules over 15 years [150]. Synthia's core algorithm utilizes a decision tree where various conditions determine the range of possible substituents or atom types. A scoring function and dynamic planning algorithm then construct complete synthetic pathways by making decisions for each inverse synthetic step, enabling the proposal of synthetic routes for all targets within 15–20 min. In 2024, López-Chávez et al. [151] used Synthia to design synthetic pathways for eight structurally diverse and synthetically challenging molecules, marking the first successful use of synthetic planning software to guide multi-step synthetic routes. The highest-scoring synthetic route was selected to synthesize the targets, achieving yields of up to 98%. Notably, the synthetic route proposed by Synthia significantly differed from the original patent-disclosed route, providing higher yields with fewer synthetic steps [152]. In recent years, AI techniques have been applied to the extraction of reaction rules. Segler et al. [153] pioneered a neural-symbolic approach to autonomously extract inverse synthesis rules from the Reaxys database without expert input. These rules were then integrated with modern Monte Carlo tree search algorithms for reaction prediction to identify the most promising inverse synthesis routes. However, template-based approaches present challenges such as high computational costs and incomplete rule coverage, which restrict their scalability [154,155].

To overcome the limitations of the template-based approach, the template-free approach draws from NLP and frames synthetic prediction, whether forward or inverse, as a Seq-2-Seq mapping problem [156]. Since molecules can be represented as SMILES strings, each chemical reaction can be encoded as a sentence, thereby treating it as a chemical language translation issue [157]. The first template-free approach to inverse synthesis analysis employed a Seq-2-Seq model, fully data-driven and trained end-to-end on a subset of experimental reactions with labeled reaction types. This model consists of a bidirectional long short-term memory (LSTM) encoder-decoder, augmented with an attention mechanism that maps the SMILES representation of reactants to those of the products [158,159]. The performance of this method has been shown to be comparable to that of a baseline rule-based expert system.

4.4. Small molecule drug design and optimization

The review discusses various scoring functions used in structure-based VS (SBVS) and suggests that comparing their

performance metrics, such as enrichment rate and selectivity, on benchmark datasets could assist researchers in selecting the most appropriate scoring function for their target proteins. Similarly, it highlights QSAR and pharmacophore-based approaches for ligand-based VS (LBVS), with performance metrics like accuracy and efficiency serving as key criteria for identifying the best method for target molecules. Additionally, the article touches on different ML and DL techniques for ADMET prediction, proposing that performance metrics such as accuracy and robustness could guide researchers in choosing the most suitable model for their compounds.

4.4.1. SBVS

SBVS, also referred to as target-based VS (TBVS), is a powerful and promising CADD method. This approach predicts the interaction between a target protein and a vast library of compounds by utilizing the 3D structure of the target. Compounds are scored and ranked based on their predicted affinity for the target's receptor binding site, facilitating the identification of those most likely to exhibit pharmacological activity against the molecular target [160].

Molecular docking, a central technique in SBVS, examines the geometric compatibility between ligands and their targets. Docking became especially valued for its low computational cost, ability to conduct virtual testing before molecular synthesis, and its effectiveness in time and cost-saving. However, while SBVS is widely used, its effectiveness can be limited by system-specific challenges. The complexity of ligand-receptor binding interactions complicates accurate parameterization, leading to difficulties in predicting binding sites and classifying compounds. This often results in high false-positive and false-negative rates [161]. Accurate SBVS depends heavily on two main components: the search algorithm and the scoring function. The search algorithm systematically explores ligand orientations and conformations at the binding site, while the scoring function predicts the binding affinity between the target and its candidate ligands. Both components are critical to the success of docking protocols [162].

Scoring functions play a critical role in molecular docking with three primary applications: i) determining the binding/alteration sites of targets and ligands, as well as the binding conformation; ii) predicting the binding affinity between proteins and ligands; and iii) optimizing potential ligands [163]. Traditionally, scoring functions are categorized into three main types: force field-based, empirical, and knowledge-based scoring functions. In recent years, ML-based scoring functions have emerged as a fourth type [164]. While traditional scoring functions are widely used, they have notable limitations, such as inadequate consideration of conformational entropy (the flexibility of the protein) and solvation energy [165]. With the abundance of experimental data available, AI algorithms can now build non-predefined, data-driven scoring functions. These models implicitly learn the eigenvectors of protein-ligand binding and their non-linear relationships with affinity. Several researchers have successfully employed ML-based scoring functions to enhance SBVS algorithms. Notable examples include random forest (RF)-Score-VS and SFCscore^{RF} based on RF, support vector regression-knowledge-based/-physico-chemical properties (SVR-KB/-EP)-score and ID-score based on support vector machine (SVM), and NNScore 2.0 and CScore based on early artificial neural networks [166–170].

While traditional ML approaches still depend on expert knowledge and feature engineering, the rise of DL algorithms offers a promising direction for scoring function modeling [171]. CNNs, for instance, can automatically extract features directly from 2D or 3D structures to predict the binding affinity between proteins and

ligands. The 3D lattices of protein-ligand structures generated by docking can serve as input to CNN models. From these lattices, relevant features, such as complex atom types, partial atomic charges, and interatomic distances, are automatically learned and extracted. These features are then used to build regression models for predicting affinity or classification models for predicting binding or non-binding interactions. CNN-based models have shown better predictive performance than traditional docking methods [172]. The introduction of DL techniques, particularly CNNs, has revitalized SBVS. Traditional scoring function approaches use pre-defined theories to design functions based on linear relationships. In contrast, AI techniques can implicitly capture intermolecular binding interactions that are challenging to model explicitly. While DL-generated scoring functions may not always outperform established ML methods, further optimization of training efficiency and interpretability is underway. Despite this, the incorporation of DL has already led to significant improvements in existing docking tools. As a result, SBVS, powered by AI and DL, is expected to become one of the most promising techniques in the drug discovery process in the near future [173].

4.4.2. LBVS

LBVS operates on the premise that structurally similar compounds exhibit comparable biological activities. Commonly employed methods in LBVS include QSAR, pharmacophore modeling, and structural similarity matching [174]. The QSAR model, developed over the past half-century, establishes a mathematical correlation between a compound's molecular properties (e.g., polarity, lipophilicity, electrical and spatial characteristics, or specific structural features) and its biological activity indicators (such as receptor affinity, inhibition constants, or rate constants). A refinement of QSAR, 3D-QSAR, directly derives binding affinities from the 3D structure of the compound, while comparative molecular field analysis (CoMFA) is a pivotal method for 3D conformational analysis. The 3D pharmacophore model involves the conformational analysis and molecular stacking of a series of active compounds to identify critical moieties that influence their activity. In pharmacophore-based VS, 3D pharmacophores, derived from active ligands, target-ligand complexes, or protein structures, are screened against a virtual compound library, retrieving molecules that satisfy the pharmacophore's criteria [175–177]. Structural similarity matching identifies compounds with analogous activities or mechanisms by comparing molecular descriptors or fingerprints.

AI-based VS models leverage molecular descriptors derived from physicochemical properties and/or topological fingerprints to build regression or classification models of activity. This approach offers greater flexibility in LBVS, eliminating dependency on program-specific functionalities. Bayesian algorithms, SVM, RF, and artificial neural networks have been extensively used to construct QSAR models, driving numerous successful applications in LBVS [178]. DNNs have outperformed traditional ML methods like Bayesian, RF, and SVM in predictive accuracy. Multi-task DNNs have further enhanced performance, with applications on 200 distinct targets for large-scale screening. The DeepTox method [179], a multi-task DNN-based toxicity prediction model, triumphed in the 2014 Tox21 dataset challenge, which involved predicting compound toxicity using a dataset of 12 high-throughput assays across 12,000 compounds [180]. The introduction of GNNs in molecular prediction allows for a more comprehensive and generalized representation of molecules, aiding in the automatic extraction of relevant molecular features for predictive modeling [181]. Recently, molecular prediction models for antimicrobial activity based on the message passing neural network (MPNN) identified eight novel

antimicrobial molecules from a database of over 107 million compounds, including the discovery of halicin, a new antibiotic that inhibits *Escherichia coli* (*E. coli*) growth [182].

The integration of pharmacophore concepts with AI techniques is still evolving, with future research likely focusing on using pharmacophore features as molecular descriptors for AI models or employing AI methods to generate pharmacophores from extensive datasets. For example, when Pharm-IF, a pharmacophore-based interaction fingerprint, was used as input to several ML algorithms for ranking small molecule docking poses, the SVM-based model outperformed other algorithms and docking scores in terms of enrichment rates [183]. It is anticipated that AI algorithms, in conjunction with increasingly refined molecular characterization methods, will soon dominate LBVS techniques.

4.4.3. Predicting pharmacogenicity

ADMET properties are essential indicators in determining whether a small molecule can develop into a viable drug, addressing key pharmacokinetic and toxicological concerns, such as the drug's ability to be effectively absorbed and reach the target tissue. Many clinical trial failures are attributed to inadequate ADMET profiles in drug candidates. Early-stage ADMET evaluation can effectively mitigate safety and efficacy concerns, thereby increasing the success rate of drug development [184]. However, traditional experimental methods for ADMET evaluation are costly and time-consuming, limiting early insights into compounds and delaying further biological validation. The advancement of computational technology, cheminformatics, and the accumulation of experimental drug data have facilitated the development of ADMET prediction models using ML and DL. These models can learn the relationship between chemical structures and pharmacokinetics from ADMET data, helping medicinal chemists avoid exploring suboptimal chemical spaces and enabling the identification of promising drug candidates [185].

ADMET prediction is a critical component of drug discovery and development. Leading global institutions and companies are increasingly integrating traditional wet lab experiments with computational methods to aid in the analysis of ADMET profiles, resulting in the development of numerous computer-aided ADMET software, databases, and online services [184]. For instance, the QikProp module from Schrödinger software predicts key pharmacokinetic parameters such as logP, logS, Caco-2 cell permeability, serum protein binding, and human ether-a-go-go related gene (hERG)-K ion channel blocking. GastroPlus, widely adopted by regulatory authorities like the U.S. FDA and the China National Medical Products Administration (NMPA), forecasts pharmacokinetic parameters including physicochemical properties, absorption, distribution, metabolism, and drug behavior after ocular and pulmonary administration [186]. The SwissADME molecular modeling team at the Swiss Bioinformatics Institute calculates physicochemical descriptors to predict pharmacokinetic properties such as oral bioavailability, blood-brain barrier permeability, and interactions with metabolic enzymes, in addition to offering a suite of widely used online tools for ADMET prediction [187]. ADMETlab 3.0 utilizes MFPs like MAS and ECFPs to train ML models such as RF, SVM, and naive Bayes for the classification and regression prediction of various ADMET properties. Additionally, ADMET SAR employs MACCS fingerprints to train SVM models, achieving superior prediction performance in 22 classification tasks, with the adoption by DrugBank, a prominent drug database. Although ML-based tools are the most widely used, employing MFPs and descriptors as features can lead to significant loss of molecular structural information, potentially limiting the predictive accuracy of these models [188,189].

DL-based ADMET prediction methods are capable of autonomously extracting feature representations from input data to model more intricate relationships. As demonstrated in the 2020 Kaggle Competition, DNNs outperformed RF models by an average of 10% across 15 large analytical datasets [190]. Researchers from leading pharmaceutical companies such as Vertex Inc., Eli Lilly & Co., and Bayer AG have also reported that DNNs either match or slightly surpass traditional ML models when trained on proprietary ADMET datasets [191–194]. The recent rise of GNNs has introduced a new paradigm in ADMET model design. GNNs represent molecules as graph structures and, through data-driven training, convert molecular structural information into low-dimensional continuous vectors, offering a more informative and compact representation compared to traditional high-dimensional sparse MFPs [195,196]. The efficacy of GNN models in predicting drug properties has been validated by frameworks such as Molecule-Net and Chemi-Net. Chemi-Net, a fully data-driven, domain-knowledge-free deep GCN developed in collaboration with Amgen, outperformed Amgen's Cubist ML program across 13 datasets in large-scale ADME property prediction, demonstrating its superior predictive accuracy and potential to accelerate drug discovery [197]. Additionally, GNNs can leverage interpretable methods, such as self-attention-based message-passing neural network (SAMPN), a message-passing neural network based on a self-attention mechanism. SAMPN has been shown to outperform both conventional GNNs and RF models in predicting lipophilicity and water solubility, while also enabling the visualization of atomic contributions to the predicted properties through attention coefficients [198].

Despite these advancements, the application of ML in ADMET prediction remains limited by the scope of publicly available training datasets. Although certain AI models have shown promise in predicting ADMET and activity properties, a critical challenge lies in the scarcity of data and the potential lack of generalizability of data-dependent models. Furthermore, these methods often focus on the similarity of physicochemical properties among approved drugs without fully accounting for their behavior within biological systems, such as permeability and clearance rates. As a result, a single predictive score often fails to capture the full complexity of drug properties, restricting its utility for guiding compound optimization.

4.5. Accelerating clinical trials

Despite promising advances in systems biology and the increased availability of high-throughput biological data, the pharmaceutical industry continues to face a decline in R&D efficiency. Clinical trial failure rates, particularly in oncology and other disease areas, can reach as high as 95%. These high failure rates contribute significantly to the inefficiencies and costs of drug development: bringing a completely new chemical entity to market can take between 7 and 10 years of clinical trials, at a capitalized cost ranging from \$1.46 billion to \$2.56 billion. The financial losses per failed clinical trial can range from \$800 million to \$1.4 billion, accounting not only for the trial costs but also for the losses in preclinical development. Applying AI technology to key steps in clinical trial design offers the potential to improve patient stratification, enhance recruitment efficiency, and ultimately increase the likelihood of trial success [199–202].

In vivo studies, which account for over 75% of the cost of developing new chemical entities, dominate drug development costs. Therefore, improvements in computational methods made early in drug development, although valuable, have a limited impact on reducing overall development expenses. The financial impact of failure in phase III clinical trials, which involve large

patient populations, can be catastrophic. Ideally, AI models used to predict late-stage trial outcomes, such as *in silico* clinical trials (ISCT), could significantly reduce these costs while improving overall success rates [203]. The concept of Virtual Physiological Human (VPH), first described in a white paper in 2005, envisions the development of patient-specific computer models that support clinical decision-making by forming virtual patient groups to test the safety and efficacy of new drugs and medical devices [204,205]. These virtual patient groups could complement traditional clinical trials by reducing the number of patients required and increasing the statistical power of the results, as well as suggesting clinical decisions. ISCT typically integrates physiological and pathological data across different spatial and temporal scales to generate patient-specific predictions. These predictions can inform decisions regarding diagnosis, prognosis, dose selection, and the identification of appropriate patient groups [206]. However, using ISCT to reduce or partially replace *in vivo* experiments presents significant challenges. One key hurdle is the inherent complexity of accurately and quantitatively modeling organisms. Without addressing these complexities, clinical trials alone may fail to provide sufficient structural and design insights to explain the failure of drug candidates. Furthermore, the reliability of ISCT-based predictions still requires validation [207]. As a result, current AI technologies are primarily focused on improving clinical trial success by intervening in several critical areas: linking patient genetic data, electronic health records (EHRs), medical literature, and clinical trial databases to predict clinical toxicity and trial success; improving trial design; assisting with patient-trial matching and recruitment; and monitoring patient adherence during trials [208,209].

4.5.1. Prediction of clinical trial results

DL models, when applied to the analysis of drug response and side effects, offer significant potential for predicting the outcomes of phase I/II clinical trials. By improving the prediction of clinical trial success, these models help optimize the drug development process [210,211]. A major cause of clinical trial failures is toxicity, which can often be predicted through computational models. For example, ProCTOR, a model designed to predict toxicity outcomes, combines chemical features of drugs with target-based features to distinguish between FDA-approved drugs and drugs that failed in clinical trials due to toxicity. Using a 48-feature set, including 10 molecular features, 34 target-related features (e.g., target tissue selectivity), and 4 drug-like rules, ProCTOR constructs an RF classifier that directly predicts the likelihood of a drug being toxic in clinical trials [212]. In addition to toxicity, a significant proportion of clinical trials fail for reasons other than safety, such as efficacy, strategic, and financial factors. Efficacy prediction remains highly complex, but combining *in vitro* cellular models with data on drug side effects can help forecast the success or failure of clinical trials. Insilico Medicine has developed a DNN based on pathway analysis techniques to predict drug side effects by analyzing the transcriptional changes drugs induce in cell lines. This network, built using transcriptomic data from drug-induced perturbations in cell cultures and pathway activation scores generated by the iPANDA algorithm, predicts clinical trial outcomes for 46 side effects. By leveraging these data-driven approaches, it is possible to predict the likelihood of success or failure in clinical trials more effectively [213].

4.5.2. Clinical trial design

AI technologies are increasingly being applied to enhance clinical trial design, patient stratification, recruitment, and monitoring, which ultimately improves trial efficiency and success rates. One example is the collaboration between Johns Hopkins University and the National Cancer Institute (NCI) to improve

clinical trial design for head and neck squamous cell carcinoma (HNSCC). They utilized the iPANDA pathway analysis algorithm to study transcriptomic data from 359 oral squamous cell carcinoma (OSCC) samples and 86 white spot samples (precancerous lesions). This analysis identified differentially dysregulated pathways between tumors and normal tissues, providing valuable insights into the complex signaling networks underlying HNSCC and paving the way for novel preventive, diagnostic, and therapeutic strategies [214]. The advent of immune checkpoint inhibitors in the treatment of HNSCC has created a need for more reliable characterization of the tumor microenvironment, particularly signaling pathways and genetic alterations associated with CD8⁺ T cell infiltration. In their study, researchers used RNA sequencing and 10 chemokine signatures to classify patients with HNSCC into subgroups with high and low CD8⁺ T cell infiltration (TCIP-H and TCIP-L, respectively). iPANDA was then applied to analyze differences in signaling pathways, somatic mutations, and copy number aberrations between these subgroups. The findings revealed that TCIP-H tumors are rich in immune checkpoint molecules, making them promising candidates for combination immunotherapy. This work provides a rationale for designing more effective immunotherapy strategies for HNSCC [215].

BERG, an AI biotechnology company, has developed Interrogative Biology, a platform that utilizes Bayesian AI analysis to integrate multi-omics molecular profiles with clinical health data, creating causal inference networks. This technology has been used to evaluate the phase I clinical trial of BPM31510 in 104 patients with advanced recurrent/refractory solid tumors [27]. For the first time, patient tissue samples and biological fluids were collected longitudinally, allowing for pan-omics analysis at multiple time points. This approach provided valuable biological insights into BPM31510's mechanism of action, confirming that Interrogative Biology can be used to assess disease biomarkers and develop actionable drug adverse event management plans. Such plans could include excluding patient subgroups that may experience adverse reactions or integrating preventive interventions into clinical trial designs [26].

NLP techniques have also been employed to extract information from electronic medical records (EMRs) to match patients with suitable clinical trials. IBM Watson has developed a clinical trial matching system that uses both structured and unstructured patient data from EMRs. This system creates detailed clinical profiles for patients and compares them with the eligibility criteria of available clinical trials, facilitating the optimization of clinical trial protocols and improving patient recruitment efficiency [216,217].

Despite these advancements, the limited data available from clinical trials and EMRs does not fully capture the complexity of biological systems, and the lack of interpretability raises concerns about the reliability of these systems and poses ethical risks. This has led to the growing importance of systems biology approaches that leverage medical data to offer deeper biological insights into drug candidates' mechanisms of action. For example, Insilico Medicine's iPANDA algorithm used a Microarray Analysis Quality Control (MAQC) dataset derived from paclitaxel-based neoadjuvant breast cancer therapy to identify biologically relevant pathway features. These features were successfully used to characterize patients with breast cancer based on their sensitivity to neoadjuvant therapy. Similarly, GNS Healthcare's Reverse Engineering & Forward Simulation (REFS) platform integrates various sources of patient data, such as EMRs, medical claims, next-generation sequencing, and other histological data, to build computational models. REFS applies ML to uncover hidden drivers of cancer progression and drug response, helping to identify new biomarkers and targets for disease and enabling more accurate patient stratification [218–220].

To improve patient adherence in clinical trials, innovative technologies are being employed to ensure more reliable monitoring. Traditional methods of tracking adherence, like pill counts or self-reported data, are prone to manipulation and inaccuracies. AbbVie has implemented AI-powered facial and image recognition algorithms through the AiCure mobile SaaS platform, which requires patients to record a video of themselves swallowing pills. The AI system then verifies that the correct person has taken the prescribed medication. In a study of patients with schizophrenia, adherence increased from 50% to 90% within six months, demonstrating the effectiveness of this AI-driven monitoring approach [221–223].

4.5.3. Drug redirection

The process of bringing new chemical entities to market encounters significant hurdles in terms of development time and cost. However, discovering new indications for an existing drug can substantially reduce these costs by repurposing it for different diseases. Drug repositioning, or drug repurposing, is a strategy that identifies novel therapeutic applications for an approved or investigational drug outside its original indication [224]. This approach allows repositioned drugs to enter phases II and III clinical trials more rapidly, with significantly reduced development costs, as pharmacokinetic, pharmacodynamic, and toxicity profiles are already established from earlier preclinical and phase I studies. Historically, successful drug repositioning has often stemmed from insights into drug pharmacology or retrospective clinical observations. For instance, sildenafil citrate, initially developed as an antihypertensive, was later repurposed by Pfizer for erectile dysfunction based on clinical findings, and thalidomide's use for erythema nodosum leprosum (ENL) and multiple myeloma arose from serendipitous discovery [225]. With the advancement of AIDD methodologies, such serendipitous successes can now be more systematically identified and traced.

By integrating systems biology with NLP techniques, the study of off-label drug use has become a preferred retargeting strategy for pharmaceutical companies. This approach leverages large-scale histological data and EHRs from patients to identify new drug indications [226]. BenevolentAI's cognitive system, JACS, utilizes AI tools and biomedical knowledge graphs to uncover novel connections within vast, unstructured datasets, such as disease, drug, and clinical trial information, enabling drug repurposing and facilitating the discovery of valuable new indications. In collaboration with BenevolentAI, Johnson & Johnson entered an exclusive licensing agreement for clinical-stage candidates, redeveloping bavisant, a histamine H3 receptor inverse agonist, originally intended for attention deficit hyperactivity disorder, for the treatment of extreme daytime sleepiness in Parkinson's disease, with phase II clinical trials underway [227,228]. In February 2020, following the World Health Organization's declaration of COVID-19 as a global health emergency, BenevolentAI employed knowledge mapping to rapidly identify baricitinib, initially developed by Eli Lilly for rheumatoid arthritis, as a potential treatment for COVID-19. Similarly, TwoXAR's DUMA™ platform mined multi-omic data, protein interactions, chemical structures, and clinical data to explore new uses for existing drugs, identifying exenatide and olopatadine as more effective treatments in animal models of rheumatoid arthritis [229–231].

In summary, AI technology has generated numerous impactful applications in the pharmaceutical industry. In drug development, AI can analyze patterns in chemical reactions, identify potential targets, design and screen candidate molecules, and predict drug kinetics and adverse reactions, all of which contribute to shortening the drug development cycle. However, challenges remain in predicting adverse reactions and interactions, such as data quality issues and

low prediction accuracy. Additionally, AI-generated drugs have not yet reached the market, and thus, the practical effectiveness of AI in drug development remains unproven.

5. The application challenges of AI in the pharmaceutical industry

The integration of new technologies, such as digitization in drug discovery and AI, has long been recognized as an irreversible trend. However, as outlined above, it is essential to acknowledge the significant variability in access to various resources, which leads to differing levels of maturity in AI-driven macromolecular drug development across different sectors. Given these disparities, it is clear that, at this stage of highly uneven data distribution and underdeveloped data-sharing models, AI-focused macromolecular drug discovery companies must prioritize the development of robust data asset production capabilities to establish genuine differentiation. Potential areas for data production include antibody screening (e.g., single B-cell analysis), target protein-function relationships (e.g., proteomics), target epitope structures, and peptide structures. The data production platform must be both unique and directly aligned with drug development needs.

In terms of data sharing, federated learning presents a more feasible solution compared to the complexities of establishing or participating in a decentralized autonomous organization (DAO). The formation of a data federation will likely center around data standardization and the level of digitalization within the participating companies. Collaborative efforts between two or three companies will be easier to implement in practice, with the barriers at the data level, driven by data volume or creation methods, being more reliable and sustainable.

In drug development, the fragmentation of data into isolated silos has led to substantial inefficiencies. One promising solution is federated learning, which enables “cooperative prosperity.” In this model, participants are autonomous entities, and organizations incentivize them to join through effective incentive and benefit-sharing mechanisms, a feature absent in traditional ML. Federated learning allows multiple participants to collaboratively train a model without sharing their raw data, thereby utilizing distributed data from various sources while protecting sensitive information, such as proprietary and confidential data. This decentralized paradigm is expected to significantly enhance the success rate of AIDD. Federated learning not only facilitates the collective training of a global model using participants' datasets but also supports personalized models for each participant. Personalized federated learning recognizes the unique characteristics of each data source, akin to creating models tailored for different demographic groups, such as the elderly or children. By incorporating local data characteristics, these personalized models can improve prediction accuracy, which is particularly beneficial in drug development for making more accurate, participant-specific predictions. Federated transfer learning, a technique that further extends the feature space and sample size, plays a pivotal role in this context. From a broader perspective, federated learning can be categorized into horizontal and vertical schemes. Horizontal federated learning applies when participants share the same feature space, such as molecular ECFP fingerprints. Vertical federated learning, on the other hand, caters to participants with distinct types of input features. Combining these two approaches, horizontal and vertical federated learning, into federated transfer learning enhances the ability to integrate data with shared and proprietary features from multiple parties. This combination allows for the expansion of both feature and sample spaces, ultimately improving the model's robustness. For instance, predicting clinical outcomes for candidate drugs often requires the integration of diverse data from pharmaceutical

companies, hospitals, and patients. Federated transfer learning can enable the pooling of this data in a way that preserves privacy and proprietary information, generating significant value for all stakeholders involved.

Federated learning represents a promising approach to drug discovery, offering a mechanism to leverage confidential datasets through secure, distributed training. This method addresses a critical challenge in the field, where the availability of large, diverse datasets has historically been limited. By facilitating the integration of data across multiple institutions, federated learning has the potential to enhance predictive models, which often operate in narrow contexts. The security and privacy features inherent in federated learning are particularly advantageous for drug discovery, especially when handling sensitive data such as genetic information. This approach can incentivize institutions to share their data, which is essential for creating robust, generalizable models. As data availability expands, the “small data” problem, commonly encountered in drug discovery, can be mitigated, leading to improved predictive accuracy and more insightful recommendations.

However, protecting intellectual property rights associated with algorithms remains a significant challenge, with barriers to entry being relatively short-lived. In the context of combining wet and dry experiments, AI-driven macromolecular drug development companies must possess strong algorithm development capabilities, particularly on the application side. Innovation in this area should be viewed as a continuous, systematic process that requires iterative updates. The output should be user-friendly for life scientists, ensuring seamless integration with wet experiments. Key considerations include whether the predictions made by algorithms are meaningful, interpretable, and capable of providing corrective feedback based on observed outcomes. In terms of algorithmic development and computational support, collaboration among companies and a more open approach to refining capabilities is essential.

Regarding business models, success in the drug development sector hinges on a company's deep understanding of the drug development process. Without this expertise, contract research organizations (CROs) will struggle to standardize their services, limiting their ability to expand offerings and attract new clients. In drug development, wet experiments, which take months to conduct, typically consume far more time than dry experiments, which are completed within days. A lack of understanding of the drug development process can dilute the efficiency gains from dry experiments, as the inefficiencies of wet experiments counterbalance the benefits. From the perspective of publicly traded companies, direct involvement in drug development has proven more valuable than offering CRO services. The primary reason for this is that downstream customers often lack the criteria to assess the quality of AI algorithms provided by CROs, resulting in low willingness to pay, an issue that is particularly pronounced in China.

Drug development companies require expertise in both life sciences and algorithms, with the essential task of developing methodologies that effectively bridge wet and dry experiments. This involves decisions on data set selection, model training processes that yield predictive results, and how to align those predictions with project advancement. Achieving success in AI-driven macromolecular drug development demands a deep integration of these capabilities. To accomplish this, companies must master technologies that extend beyond traditional ML algorithms and biopharmaceuticals, incorporating knowledge from fields such as synthetic biology (e.g., the synthesis of artificial proteins) and engineering automation (e.g., digital adaptation of laboratory automation).

Given the current challenges, there is significant demand within the pharmaceutical industry for advanced technologies capable of

accelerating drug discovery and validation (Fig. 4). Hundreds of collaborations between pharmaceutical companies and AI tech firms have already occurred worldwide, marking a notable shift in the pharmaceutical sector from skepticism to interest in AI. However, the question remains: how far is this shift from interest to trust? AI's potential to reshape the pharmaceutical landscape could integrate the entire AI ecosystem into pharmaceutical industry. This raises the question of whether computational and traditional pharmaceutical industry will eventually become parallel models, much like online and offline shopping, where online shopping serves as a form of VS. The future remains uncertain. One of the central challenges is whether the vast array of variables in complex biological systems can be accurately quantified and analyzed to identify novel drug targets and better assess the effects of drugs. Though one undeniable truth is that AI has the potential to extract value from data across the entire drug discovery cycle, many unknowns remain to be explored. Although data is not synonymous with science, virtually all scientific breakthroughs are identified and validated through data. As the volume of data continues to grow, drug development data is evolving into big data, and AI is currently the most effective tool for managing and extracting insights from this data.

The rapid advancement of AI technology has cemented the digitalization and AI applications in drug discovery and development as an irreversible trend. However, due to varying challenges in data acquisition, the maturity of AI-driven macromolecular drug development remains uneven across its stages. To differentiate themselves, AI drug discovery companies must establish robust data asset production capabilities. These platforms must be unique and closely aligned with the drug development process. In terms of data sharing, federated learning offers a more feasible solution than establishing or participating in a DAO, as it prioritizes data standardization and the digital maturity of participating entities. Federated learning ensures data privacy through distributed training, thereby enhancing the success rate of AIDD while enabling the creation of personalized models for participants. Moreover, AI drug development companies must foster the ability to innovate algorithmically and maintain an open, collaborative approach to further strengthen their capabilities. Business model-wise, direct involvement in drug development is likely to generate greater corporate value compared to providing CRO services. Drug development companies must also cultivate expertise in both life sciences and algorithms, developing methodologies that effectively integrate wet lab and dry lab experiments. Despite the challenges, the demand for advanced technologies that facilitate drug discovery and validation remains immense within the pharmaceutical industry, evidenced by hundreds of global collaborations. The attitude of the traditional pharmaceutical industry toward AI has shifted from skepticism to interest, though a significant gap remains between interest and trust. AI's application in drug discovery will likely integrate the entire AI ecosystem into the pharmaceutical sector. However, whether computational and traditional pharmaceutical models will evolve into parallel systems remains to be seen. Ultimately, whether AI can reshape and transform the drug discovery process by extracting value from data across its entire cycle will define the future trajectory of the industry.

6. Overcoming the challenges and potential future directions

AI, as one of the most advanced technologies today, has made significant progress across various industries. However, despite its impressive performance, AI still faces numerous limitations and challenges that hinder its deeper development and broader application. A key issue is the reliance on large volumes of high-quality data. While unsupervised learning and reinforcement learning may

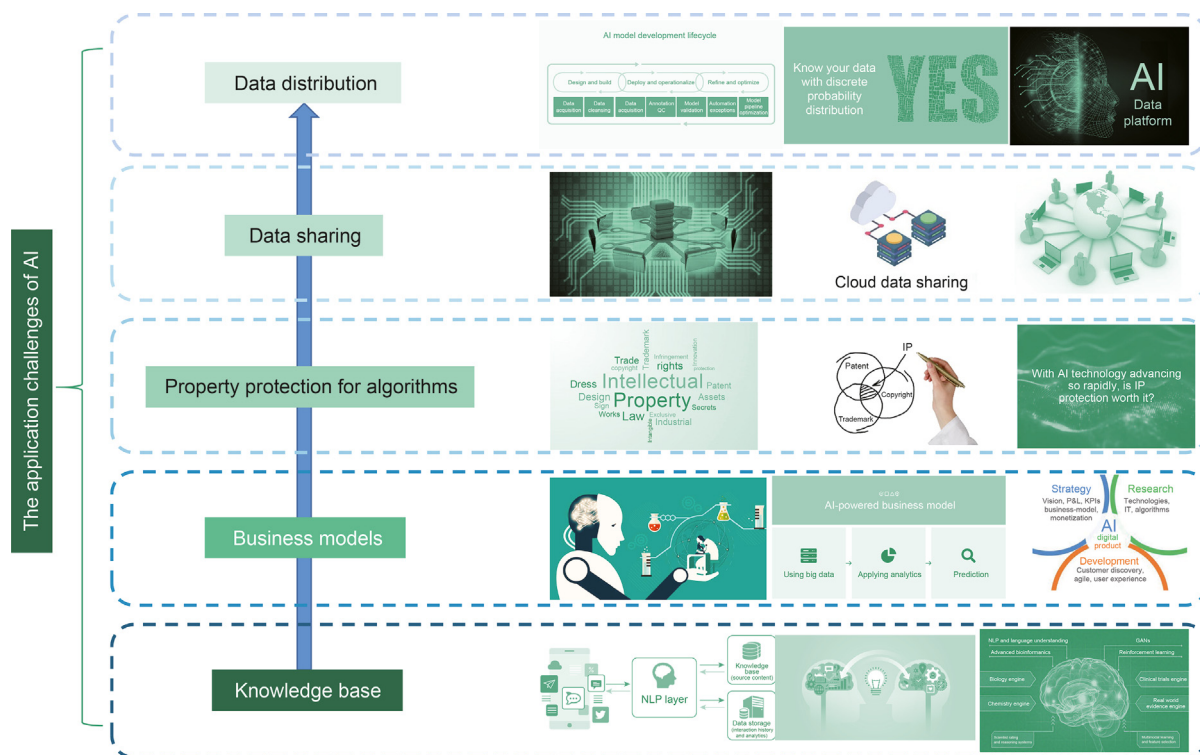


Fig. 4. Application challenges of artificial intelligence (AI) in drug research and development (R&D). QC: quality control; IP: intellectual property; P&L: profit and loss; KPIs: key performance indicators; IT: information technology; NLP: natural language processing; GAN: generative adversarial network.

reduce the demand for labeled data in certain cases, many AI applications still require extensive datasets for training. For instance, tasks such as image recognition and NLP typically necessitate millions or even billions of data samples to achieve high performance. Data quality is crucial; errors, biases, or gaps within training data can adversely impact the outputs of AI models. Although data cleansing and preprocessing are essential for ensuring data quality, these processes are often time-consuming and complex. Furthermore, the collection and use of large datasets raise concerns regarding privacy and security. Leaks or misuse of user data may result in severe violations of privacy rights. Consequently, ensuring privacy protection while utilizing data for AI applications presents a significant challenge.

Another pressing issue is the transparency and interpretability of models. Many AI models, particularly DL models, are often regarded as “black boxes” due to the difficulty in explaining their internal mechanisms. While these models can provide accurate predictions, understanding how they make decisions remains challenging, especially in high-stakes fields such as healthcare and finance. To foster trust and accountability, AI models must be interpretable. Stakeholders, including users and regulatory bodies, need to understand how decisions are made to ensure fairness and reliability. Although researchers are developing techniques to enhance transparency and interpretability, this issue remains unresolved.

Bias and fairness are also prominent concerns within DL. AI systems may inherit or even amplify biases present in their training data. For example, datasets that include biases related to gender, race, or other factors may lead to discriminatory outcomes. Even when data appears neutral, algorithms can introduce biases through inconsistent processing of data from different groups, potentially disadvantaging certain populations. Ensuring fairness in AI systems is a complex task that requires stringent control at every stage, from model design and data collection to model evaluation.

Researchers and developers are actively exploring strategies to reduce and eliminate bias, but achieving completely fair AI remains a challenging goal.

Training complex AI models demands substantial computational resources and time. For example, training a large DL model may take weeks and require immense computational power. This presents a significant challenge for resource-constrained institutions and developers. Additionally, the training and operation of AI models consume considerable energy, leading to high power consumption and a substantial carbon footprint. With the continued proliferation of AI applications, addressing the environmental impact of AI and reducing energy consumption has become an increasingly important issue. Researchers are striving to develop more efficient algorithms and hardware to diminish the computational demands and energy consumption of AI systems. Moreover, advancements in quantum computing and edge computing technologies hold promise for significantly enhancing the efficiency of AI systems in the future.

Currently, most AI systems fall under the category of narrow AI, optimized for specific tasks. For instance, an AI designed for image recognition excels in identifying images but cannot perform NLP tasks. The realization of artificial general intelligence (AGI), an AI capable of executing various tasks across different domains, remains a distant goal. Existing AI systems are limited in their understanding and reasoning capabilities. While they can process extensive datasets and conduct complex computations, they still struggle with common-sense reasoning, contextual judgment, and complex problem-solving. This restricts their effectiveness in broader applications. AGI would need to possess the ability to learn autonomously and adapt to new environments; however, current AI systems face difficulties in adapting to dynamic and changing environments, often requiring retraining or significant adjustments to handle new tasks.

The rapid expansion of AI applications also brings forth a series of ethical, social, and legal challenges, particularly in medical AI research. These challenges include the ethical use of informed consent, ensuring safety and transparency, mitigating algorithmic bias, protecting patient data privacy, and maintaining the security and efficacy of AI technologies. Issues related to accountability, intellectual property, and safeguarding AI systems from cyber threats further complicate the situation. Healthcare institutions, as research channels and ethical overseers, must comprehensively address these issues and manage the inherent ethical concerns within their AI research and applications.

The swift advancement of AI technologies has led to increasingly complex and autonomous algorithms, raising the “black box” issue where the decision-making processes of algorithms are difficult for humans to understand. This problem not only hinders the interpretability of algorithms but may also result in negative consequences, such as algorithmic collusion or abuse of power. To address these issues, explainable AI (XAI) has emerged as a critical solution, emphasizing the transparency and interpretability of algorithmic decisions [232]. XAI enhances algorithmic transparency, enabling researchers to better understand, optimize, and trust AI systems. In the context of patent disclosures for algorithms, the application of XAI becomes particularly important. XAI assists patent examiners in understanding complex algorithms, ensuring that patent disclosures are adequate and transparent. Patent protection plays a pivotal role in promoting the interpretability and transparency of AIDD models [233]. It requires inventors to disclose technical details of their algorithms, including underlying principles and decision-making processes. These requirements urge inventors to provide sufficient information, ensuring that others can understand and potentially replicate their technologies. In the field of AI, especially in drug design, patent protection places significant emphasis on the interpretability of algorithms to ensure that examiners and the public can comprehend how AI systems reach conclusions. For instance, a team by Collins and co-workers [234] at MIT utilized an interpretable DL model to identify a novel antibiotic from over 12 million compounds, effectively targeting methicillin-resistant *Staphylococcus aureus* (MRSA). This success story underscores the importance of interpretability in AIDD, as it enables researchers to understand how the model makes predictions, thereby improving the design of more effective therapeutic agents. The drug developed using AI has progressed to clinical trials, underscoring how interpretability builds trust between researchers and clinicians by clarifying drug mechanisms and accelerating development. The mandatory disclosure requirements of patent protection significantly enhance the interpretability and transparency of AIDD models. These success stories indicate that interpretability is not only a technical necessity but also a crucial factor in gaining societal trust and facilitating successful commercialization.

In addition to the aforementioned content, new methods continue to emerge in the field of XAI, bringing new opportunities for drug development. For example, Chemical-explainable GNN (ChemXGNN), as a cutting-edge XAI approach, exhibits unique advantages in drug discovery [235]. It combines the powerful molecular structure representation capabilities of GNN with interpretability techniques, enabling in-depth analysis of drug-target interactions at the molecular level. When predicting the affinity of a specific drug for a target, ChemXGNN not only provides prediction results but also demonstrates the molecular structural features that the model focuses on during decision-making through its interpretability module, such as specific chemical bonds and functional groups. This clear explanation of the molecular structure-activity relationship aids medicinal chemists in understanding the rationale behind model decisions, allowing for more

targeted optimization of drug molecular structures and accelerating the new drug development process.

While AI offers significant advancements across various fields, its widespread adoption raises concerns related to ethics, social implications, and employment. AI has the potential to automate certain professions, which may lead to job losses and social challenges. Although AI also creates new employment opportunities, ensuring a balance between supply and demand in the job market to avoid mass unemployment remains a critical social issue. Additionally, the application of AI in sensitive areas such as military and surveillance raises ethical and moral dilemmas. For instance, the use of automated weapons and mass surveillance systems may violate human rights and privacy, raising concerns about the responsible use of AI technologies. As AI technology rapidly evolves, existing legal and regulatory frameworks often struggle to keep pace. Governments and international organizations must collaborate to develop and implement effective laws and regulations to govern the development of AI and ensure the protection of public interests.

The security and privacy of medical data form the foundation for advancements in AI within the pharmaceutical sector. To meet the growing demand for medical data, the development of advanced data protection technologies is crucial. Simultaneously, efforts are being made to enhance the interpretability and reliability of AI models, thereby bolstering public and regulatory confidence in AI-driven pharmaceuticals. Improving data quality and sample representativeness is essential for enhancing the reliability and generalization capabilities of AI models. This requires strict enforcement of standards and regulations regarding data collection and processing. Clearly defining the responsibilities and rights of all stakeholders, protecting intellectual property, and adhering to ethical standards in medical research and clinical trials are paramount. These measures include formulating and enforcing relevant laws and regulations, as well as establishing ethical guidelines. Early collaboration between companies and regulatory agencies is vital for ensuring the compliance of AI models and enhancing their credibility. Regulatory bodies should establish frameworks and standards to facilitate this collaboration. Furthermore, verification plans must be developed to validate the robustness of AI models and ongoing assessments to ensure their effectiveness and reliability.

The application of AI in drug development holds immense promise but faces several challenges. Key issues include data quality and accessibility, as biases in training datasets may lead to unfair AI outcomes, while data silos hinder the sharing and utilization of critical information. Transparency and interpretability of models remain significant obstacles, as DL models typically operate as “black boxes”, making it difficult to understand their internal workings, raising concerns about the reliability and fairness of AI-driven predictions and decisions. Additionally, the generalization and robustness of AI models require further improvement to mitigate risks of overfitting and adversarial attacks. The substantial computational resources and energy consumption required for training and deploying complex AI models also present environmental and logistical challenges. Moreover, the integration of AI into healthcare introduces ethical, social, and legal issues, including privacy and security, algorithmic bias, fairness, and the establishment of accountability and regulatory frameworks. Despite these challenges, the potential of AI in drug discovery remains vast, necessitating ongoing research, collaboration, and ethical oversight to ensure responsible usage and improve patient outcomes.

In the realm of ADMET prediction, new platforms continue to emerge with ongoing technological advancements. For instance, admetSAR3.0 [236] has been optimized and expanded based on existing ADMET prediction platforms. Compared to previous

versions, admetSAR3.0 updates a substantial amount of experimental data and constructs more complex and accurate predictive models, significantly enhancing the accuracy and reliability of predicting various ADMET properties of drugs. It employs more advanced ML algorithms capable of better handling complex molecular structure data and uncovering potential relationships between chemical structures and ADMET properties. Additionally, admetSAR3.0 offers a more user-friendly interface and richer functionalities, allowing predictions for individual ADMET properties while supporting joint predictions and analyses of multiple properties, providing drug developers with comprehensive information to more accurately assess the drug-likeness of candidate compounds in the early stages of drug development, thereby enhancing development efficiency and reducing costs.

The integration of AI within the pharmaceutical industry will increasingly prioritize data privacy protection, utilizing technologies such as federated learning to securely integrate data across institutions, thereby enhancing the predictive capabilities of drug discovery models. With the advancement of personalized federated learning, AI will be able to develop customized models tailored to individual participants, facilitating more personalized drug development and treatment strategies.

Moreover, it is noteworthy that large language models are gradually emerging in the field of drug development. With their powerful language understanding and generation capabilities, large language models are subtly transforming the traditional drug discovery process [237]. For instance, during the drug target discovery phase, large language models can rapidly analyze and comprehend vast amounts of biomedical literature, extracting potential drug target information. Previously, researchers needed to spend considerable time reviewing literature to identify disease-relevant potential targets, while large language models can process and synthesize this information in a short period, uncovering new target associations and improving target discovery efficiency [238]. In the drug design phase, large language models can understand and generate textual descriptions related to chemical structures, assisting in the design of novel drug molecules. They can propose molecular structures with potential activity based on known drug activity and structural relationships, providing medicinal chemists with more design ideas and broadening the possibilities for drug design, thus enhancing the efficiency and accuracy of drug design. Additionally, large language models can participate in the design and evaluation of clinical trials, providing suggestions for optimizing clinical trial protocols by analyzing extensive clinical data and research literature, helping to identify more reasonable trial endpoints, sample sizes, and patient inclusion criteria, thereby improving the quality and success rate of clinical trials.

7. Conclusion

The pharmaceutical industry is currently experiencing exponential growth in data, and the most effective AI approaches in modeling do not rely solely on pure AI processes. In fact, the synergy between humans and AI often surpasses the capabilities of either independently. Much like in the realm of chess, where human-machine collaboration can outperform either humans or computers acting alone, the integration of human insights with AI technologies can yield superior outcomes. AI methodologies require systematic organization and development and attention, exploration, and experimental application across various fields can accelerate the maturation and innovation of AI technologies. As the cycle of “big data → more precise models → better drugs → more and better data” gradually matures in practice, AI-driven advancements in pharmaceuticals will be significantly accelerated. However, the widespread application and integration of any technology require time, and its

development proceeds in a wave-like manner. Before AI and data-driven pharmaceutical models can fully realize their potential, further exploration and practical application are necessary. In summary, this review aims to promote the application of AI in drug discovery, address current challenges, encourage collaborative efforts among stakeholders, ensure the protection of intellectual property, and outline a future blueprint in which AI plays a crucial role in advancing pharmaceutical research. It calls upon the pharmaceutical community to leverage the powerful capabilities of AI, with the ultimate goal of improving patient outcomes through more efficient and effective drug discovery.

It is noteworthy that despite the tremendous potential of AI in the pharmaceutical sector, several limitations persist. Data bias issues are particularly prominent. The training datasets may exhibit various biases, such as sample selection bias and measurement bias. These biases can result in unfair outcomes from AI models, affecting their predictive accuracy and reliability. For instance, in drug target prediction, if the training dataset contains an excess of target data related to a particular disease while having insufficient data for others, the model may predict targets for the data-rich disease more accurately, while predictions for the data-scarce diseases may be biased, thereby impacting the specificity and efficacy of new drug development.

Model interpretability presents another significant challenge. Many AI models, especially DL models, are often regarded as “black boxes.” For example, while DL models used for drug design can predict the activity of compounds, they struggle to explain the basis of their predictions and the internal decision-making processes. This lack of transparency makes it difficult for researchers to comprehend model behavior and assess the reliability and safety of these models, limiting their application and promotion in practical drug development.

Ethical issues cannot be overlooked either. The application of AI in the pharmaceutical field involves the handling of vast amounts of patient data, which includes personal privacy information. Any leakage or improper use of this data could severely infringe on patients' privacy rights. Additionally, in clinical trials, when AI models are employed for patient stratification and selection, any biases in the models may lead to the unfair exclusion of certain patient groups from trials, thereby affecting the fairness and accessibility of drug development. Furthermore, disputes exist regarding the ownership of patents generated by AI and the delineation of responsibilities, necessitating further legal and ethical guidelines for clarification.

In conclusion, while the development prospects for AI in the pharmaceutical field are promising, achieving its widespread application and in-depth development necessitates attention to and resolution of these limitations. Moving forward, there is a need to enhance data quality management, develop interpretable AI models, and establish robust ethical and legal frameworks to promote the healthy and sustainable development of AI in the pharmaceutical sector.

CRediT authorship contribution statement

Chen Fu: Writing – original draft, Software, Project administration, Funding acquisition, Data curation, Conceptualization.
Qiuchen Chen: Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that there are no conflicts of interest.

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